

Two theoretical limits of the underexpanded-jet campaign:
 (i) the NPR lower bound, and (ii) why the axisymmetric (RZ)
 model
 is valid only in the near field — a vorticity-equation argument

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2 July 2026

1 The lower bound of NPR: the choking threshold

For a convergent (sonic) nozzle discharging into quiescent ambient at p_∞ , the jet is *underexpanded* only if the nozzle is choked. Isentropic flow gives the exit static pressure of a choked nozzle, $p_e/p_0 = \left(\frac{2}{\gamma+1}\right)^{\gamma/(\gamma-1)}$, so the exit is exactly pressure-matched ($p_e = p_\infty$) when

$$\text{NPR} \equiv \frac{p_0}{p_\infty} = \left(\frac{\gamma+1}{2}\right)^{\frac{\gamma}{\gamma-1}} = 1.892\,93 \quad (\gamma = 1.4). \quad (1)$$

This is the “1.8xx” bound quoted in the literature (e.g. Franquet *et al.*, *Prog. Aerosp. Sci.* 2015):

- NPR < 1.893: nozzle unchoked, exit subsonic, ordinary subsonic jet — no shock-cell train exists at all;
- NPR = 1.893: perfectly expanded sonic jet ($p_e/p_\infty = 1$, $M_j = 1$);
- NPR > 1.893: underexpanded sonic jet, with the fully-expanded Mach number $M_j = \sqrt{\frac{2}{\gamma-1} \left[\text{NPR}^{(\gamma-1)/\gamma} - 1 \right]}$.

Our campaign uses $\text{NPR} = p_0/p_\infty$ directly, so the current case matrix sits as follows:

NPR	p_e/p_∞	M_j	regime
1.893	1.000	1.000	threshold (perfectly expanded)
2	1.057	1.046	+5.7% above the limit — weakest possible wave train
3	1.585	1.358	moderately underexpanded, regular reflection
3.8	2.0	1.53	empirical Mach-disk onset ($p_e/p_\infty \simeq 2$)
4–5	2.1–2.6	1.56–1.71	Mach disk established

Thus **NPR = 2 is essentially at the lower boundary of the underexpanded regime**: the exit over-pressure is only 5.7%, the wave train consists of weak isentropic compression/expansion waves in regular reflection (no embedded shocks strong enough for a Mach disk), and all density gradients are smooth — which is also why gradient-based AMR tagging self-limits for this case.

2 Why RZ-axisymmetry is a near-field model: the vorticity equation

2.1 Setup

Compressible vorticity transport (curl of the momentum equation):

$$\frac{D\boldsymbol{\omega}}{Dt} = \underbrace{(\boldsymbol{\omega} \cdot \nabla) \mathbf{u}}_{\text{stretching/tilting}} - \underbrace{\boldsymbol{\omega}(\nabla \cdot \mathbf{u})}_{\text{dilatation}} + \underbrace{\frac{\nabla \rho \times \nabla p}{\rho^2}}_{\text{baroclinic}} + \underbrace{\nabla \times \left(\frac{1}{\rho} \nabla \cdot \boldsymbol{\tau} \right)}_{\text{viscous}}. \quad (2)$$

In cylindrical coordinates with the axisymmetric, swirl-free ansatz of the RZ solver ($\partial_\theta \equiv 0$, $u_\theta \equiv 0$):

$$\mathbf{u} = u_r \mathbf{e}_r + u_z \mathbf{e}_z, \quad \boldsymbol{\omega} = \nabla \times \mathbf{u} = \omega_\theta \mathbf{e}_\theta, \quad \omega_\theta = \frac{\partial u_r}{\partial z} - \frac{\partial u_z}{\partial r}. \quad (3)$$

All vorticity is azimuthal: vortex lines are perfect circles around the axis (ring vortices), by construction and forever.

2.2 The stretching term collapses to ring dilation

Evaluate the 3-D production term under the ansatz. Since the scalar components are θ -independent but $\partial_\theta \mathbf{e}_r = \mathbf{e}_\theta$,

$$(\boldsymbol{\omega} \cdot \nabla) \mathbf{u} = \frac{\omega_\theta}{r} \partial_\theta (u_r \mathbf{e}_r + u_z \mathbf{e}_z) = \frac{u_r \omega_\theta}{r} \mathbf{e}_\theta. \quad (4)$$

The tensorial stretching $(\boldsymbol{\omega} \cdot \nabla) \mathbf{u} = \mathbf{S} \boldsymbol{\omega} + \frac{1}{2} \boldsymbol{\omega} \times \boldsymbol{\omega} \dots$ degenerates into the single scalar u_r/r : the only way an axisymmetric vortex ring can “stretch” is by increasing its own radius. Substituting (4) into (2) and using continuity $D\rho/Dt = -\rho \nabla \cdot \mathbf{u}$ together with $Dr/Dt = u_r$ gives the scalar transport law

$$\boxed{\frac{D}{Dt} \left(\frac{\omega_\theta}{\rho r} \right) = \frac{1}{\rho^3 r} (\nabla \rho \times \nabla p)_\theta + \frac{1}{\rho r} \left[\nabla \times \left(\frac{1}{\rho} \nabla \cdot \boldsymbol{\tau} \right) \right]_\theta} \quad (5)$$

i.e. **for inviscid, barotropic flow $\omega_\theta/(\rho r)$ is a material invariant** (the compressible generalisation of the classical axisymmetric invariant ω_θ/r).

2.3 Consequences: no amplification, no cascade

(a) **Algebraic bound on vorticity.** Integrating (5) along a particle path,

$$\omega_\theta(t) = \omega_\theta(0) \frac{\rho(t) r(t)}{\rho(0) r(0)} + (\text{baroclinic integral}). \quad (6)$$

Both ρ and r are bounded, so vorticity can grow only *algebraically*, through ring dilation or compression. Contrast 3-D: writing $\boldsymbol{\omega} = \omega \mathbf{e}_\omega$, the stretching term gives $D\omega/Dt = (\mathbf{e}_\omega \cdot \mathbf{S} \mathbf{e}_\omega) \omega + \dots$, and the well-known preferential alignment of $\mathbf{e}_\omega$ with the extensional strain eigenvector makes the growth rate positive on average — *exponential* amplification, terminated only at the Kolmogorov scale. This is the engine of the direct energy cascade; RZ removes it identically.

(b) Enstrophy budget. The inviscid enstrophy production in 3-D is $\int \boldsymbol{\omega} \mathbf{S} \boldsymbol{\omega} dV$, positive on average ($\propto -\langle S^3 \rangle$, velocity-derivative skewness $\simeq -0.4$ in real turbulence). Under the RZ ansatz

$$\boldsymbol{\omega} \mathbf{S} \boldsymbol{\omega} = \omega_\theta^2 S_{\theta\theta} = \omega_\theta^2 \frac{u_r}{r}, \quad (7)$$

which is sign-indefinite and tied to the *mean* radial motion — there is no self-amplification mechanism. Enstrophy is (up to baroclinicity and viscosity) a Casimir-constrained quantity, exactly as in 2-D turbulence.

(c) Wrong cascade direction. With stretching absent, the (r, z) -plane dynamics possess two quadratic inviscid invariants (kinetic energy and the family $\int \rho f(\omega_\theta/\rho r) dV$). By the standard Fjørtoft/Kraichnan argument this forces a *dual cascade*: energy flows to *large* scales (vortex pairing and merging), enstrophy to small scales. The phenomenology in our RZ solutions — shear-layer rings that pair, grow, and form large coherent “puffing” structures instead of breaking down — is not a numerical artefact but the *correct* solution of the axisymmetric equations, which are simply the wrong equations for a turbulent far field. A real jet obeys the single-invariant 3-D Richardson–Kolmogorov direct cascade.

(d) Modal truncation. Decomposing any disturbance as $\sum_m \hat{q}(r, z) e^{im\theta}$, the RZ solver retains only $m = 0$. Linear stability theory of round jets (Batchelor & Gill 1962; Michalke 1971) shows: for the thin annular shear layer near the lip ($\delta_\omega/R \ll 1$) the $m = 0$ and $|m| = 1$ modes have comparable growth rates — so the *initial* roll-up into rings is genuinely captured; but for the thick, developed profile downstream the *only* unstable mode is the helical $|m| = 1$, and the dominant secondary instabilities of the ring street (Widnall-type azimuthal core waves, elliptic instability of the cores, braid streamwise vortices) are all $m \neq 0$. The RZ model therefore *cannot even represent linearly* the instabilities that govern transition and far-field dynamics.

2.4 Why the near field survives

The same decomposition explains the region of validity:

1. **Wave-system gasdynamics is $m = 0$ physics.** The Prandtl–Meyer fan, intercepting/barrel shock, Mach disk and reflected shock are steady solutions of the axisymmetric Euler equations (method of characteristics + Rankine–Hugoniot); the real jet’s near-field wave train *is* axisymmetric to leading order.
2. **Near-field vorticity is baroclinic, and its $m = 0$ part is retained.** Equation (5) keeps the full axisymmetric baroclinic source $(\nabla \rho \times \nabla p)_\theta$: vorticity generated at curved shocks and at the Mach-disk slip line is captured — these set the shear layers that bound the first cells.
3. **Initial K–H roll-up is quasi-planar.** With $\delta_\omega \ll R$ the annular layer behaves locally like a plane mixing layer whose primary instability is two-dimensional (Michalke); rings are indeed observed in real jets for the first 1–2 pairings.
4. **Failure begins at the first 3-D secondary instability.** After $\mathcal{O}(1)$ pairings the Widnall instability grows on the ring-turnover timescale; beyond that station stretching, azimuthal modes and the direct cascade — all absent in RZ — control the physics.

Empirically, in this campaign (NPR = 3): centerline and field data agree between RZ and full 3-D to $< 3\%$ up to $z \approx 1.8 D_e$ (through the first shock cell and Mach disk), while at $z = 20 D_e$ the RZ centerline velocity is $1.8\times$ the 3-D value and the high-NPR RZ far field exhibits 25–40% rms coherent puffing — precisely the inverse-cascade signature predicted above.

Summary. RZ axisymmetry deletes exactly the two ingredients that make jet turbulence three-dimensional — exponential vortex stretching (replaced by the material invariant $\omega_\theta/\rho r$, Eq. 5) and all $m \neq 0$ azimuthal modes — while retaining everything that fixes the near-field wave structure (axisymmetric gasdynamics and $m = 0$ baroclinicity). It is therefore a quantitatively reliable *near-field* model (first few shock cells) and a qualitatively wrong *far-field* model, independent of grid resolution.